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Review

# Role of platelet in Parkinson's disease: Insights into pathophysiology & theranostic solutions

Samir Kumar Beura, Abhishek Ramachandra Panigrahi, Pooja Yadav, Sunil Kumar Singh *

*Department of Zoology, School of Biological Sciences, Central University of Punjab, Ghudda, Bathinda, Punjab 151401, India*

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ABSTRACT

Parkinson's disease (PD) is the second-most-common neurodegenerative disease characterized by motor and non-motor dysfunctions, which currently affects about 10 million people worldwide. Gradual death and progressive loss of dopaminergic neurons in the pars compacta region of substantia nigra result in striatal dopamine deficiency in PD. Specific mutation with further aggregation of α-synuclein in the intraneuronal inclusion bodies is considered the neuropathological hallmark of this disease. PD is often associated with various organelle dysfunctions inside a dopaminergic neuron, including mitochondrial damage, proteasomal impairment, and production of reactive oxygen species, thus causing subsequent neuronal death. Apart from several genetic and non-genetic risk factors, emerging research establishes an association between cardiovascular diseases, including coronary heart disease, myocardial infarction, congestive heart failure, and ischemic stroke with PD. The majority of these cardiovascular diseases have an origin from atherosclerosis, where endothelial dysfunction following thrombus formation is significantly regulated by blood platelet. This non-nucleated cell fragment expresses not only neuron-specific molecules and receptors but also several PD-specific biomarkers such as α-synuclein, parkin, PTEN-induced kinase-1, tyrosine hydroxylase, dopamine transporter, thus making platelet a suitable peripheral model for PD. Besides its similarity with a dopaminergic neuron, platelet structural alterations, as well as functional abnormalities, are also evident in PD. However, the molecular mechanism behind platelet dysfunction is still elusive and quite controversial. This state-of-the-art review describes the detailed mechanism of platelet impairment in PD, addressing the novel platelet-associated therapeutic drug candidates for plausible PD management.

## 1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder (NDD) that affects the pars compacta region of substantia nigra (SNpc) containing dopaminergic (DA) neurons, thus causing motor as well as non-motor impairments (Bloem et al., 2021). The first narration about PD originates back in 1817 when a British physician named James Parkinson described a malady as *Paralysis Agitans* (Shaking Palsy) in his text "Essay on the Shaking Palsy" (Tysnes and Storstein, 2017). Later, a famous French neurologist of the 19th century, Jean-Martin Charcot, studied this illness and renamed *Paralysis Agitans* as "Maladie De Parkinson" or "Parkinson's Disease" between 1868 and 1881, and the term ensues till-date (Blonder, 2018). The selective degeneration of the distinct DA neurons in the SNpc region is caused by the accumulation of Lewy bodies containing intracellular aggregates of α-synuclein protein (Bloem et al., 2021). The α-synuclein, and other associated factors, result in protein misfolding along with their aggregation, mitochondrial damage causing energy failure, generation of oxidative stress, and impairment of protein clearance pathways, which are proposed to be the primary reasons behind the death of DA neurons (Fig. 1) (Kalia and Lang, 2015). Although the complete etiology of this disease is obscure, it has been proposed that environmental factors, including pesticide exposure, heavy metal use, and drug abuse, greatly influence the time and severity of this disease onset. In contrast, genetic contribution partly determines the lethality of this disease (Ball et al., 2019). Apart from these involvements, several cardiovascular diseases (CVDs), including ischemic stroke, myocardial infarction, coronary heart disease, and congestive heart failure, are considered promising risk factors for the development of PD in recent times (Park et al., 2020). Thrombosis associated with atherosclerotic plaque formation is regarded as the principal reason behind the majority of CVDs, where platelet, a blood component, plays a pivotal role in regulating the formation as well as

* Corresponding author.
*E-mail addresses:* singh.sunil06@gmail.com, sunil.singh@cup.edu.in (S.K. Singh).

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stability of vascular thrombi (Koupenova et al., 2017).

Platelets are very small non-nucleated cell fragments having a diameter of 2–4 μm that originate from megakaryocytes in bone marrow *via* thrombopoiesis. Platelets are the smallest blood cells having discoid shapes, which remain in blood circulation for 7–10 days, and later get destroyed by phagocytosis in the liver and spleen (van der Meijden and Heemskerk, 2019). An average healthy adult can produce about $10^{11}$ platelets per day, whereas the average platelet count in circulation ranges from 150,000 to 400,000 per microliter of blood (Ghoshal and Bhattacharyya, 2014). Owing to their unique granular composition, platelets help in hemostasis, thrombosis, tissue proliferation, and immunomodulation (Fig. 2) (van der Meijden and Heemskerk, 2019). In addition to their conventional roles, platelets are nowadays gaining significant attention for their outstanding contributions in providing intercellular communication between the brain and blood, thus considering them the peripheral models for NDDs (Leiter and Walker, 2020). Furthermore, platelets also store and secrete several neuron-specific molecules, including neurotransmitters like glutamate, dopamine, serotonin, and histamine, as well as express their transporter and receptors besides tau, brain-derived neurotrophic factor, monoamine oxidase, amyloid-beta, and reelin (Fig. 4) (Beura et al., 2022). Although there are significant research findings available in deciphering the involvement of platelet in PD, there is no thorough review that exists yet addressing the role of platelet in the pathophysiology of PD. As the platelet structural and functional alterations are evident in PD, here in this concise review, we appraise the detailed role of platelet in the pathoprogression of PD (Fig. 3). Being a rich source of growth factors, we have highlighted the utilization of several platelet-rich growth factors for DA neurogenesis in PD. However, we are also addressing the novel platelet-based diagnostic and therapeutic solutions for the efficient management of PD. Further experiments could unveil a mechanistic insight into the molecular signaling of other PD-specific proteins in platelet, which will help in proposing additional platelet-associated medications to be used for plausible therapeutic treatment in PD.

## 2. Parkinson's disease (PD)

PD is a common neurodegenerative disorder that can cause significant disability and decreased quality of life, where its pathogenesis and origin remain essentially controversial. In most instances, PD is considered to result from a complex interaction between several genetic and multiple environmental factors, though rare monogenic forms of this disease also exist (Poewe et al., 2017). The various forms of PD generate an alarming health care issue with a profound global impact. The degeneration of the nigrostriatal dopaminergic pathway involves various neuronal systems, causing several neuromediator dysfunctions that account for the complex patterns of functional deficits in PD (Bloem et al., 2021).

### *2.1. Epidemiology and symptoms of Parkinson's disease*

The epidemiological investigations in PD have provided several vital pieces of evidence about the behavioral and environmental patterns of the disease pathogenesis as well as progression. The current global prevalence of this disease is around 0.3 %, which increases very sharply to about 3 % in those of 80 years or older. It had already affected approximately 6.1 million people worldwide, according to an estimation in 2016, and the number of disease cases is a matter of concern, as the number of affected people with PD is expected to double by 2030 (Poewe et al., 2017). The annual incidence rate of PD is determined to be 14 in every 1,00,000 individuals in the total population and 160 PD-affected persons out of 1,00,000 people with an age of 65 years or older (Ascherio and Schwarzschild, 2016). Although most of the PD cases are restricted to an age of 65 years or more, the age of onset for almost 25 % of PD-affected individuals is below 65 years (Bloem et al., 2021). Moreover, in a report, it has been found that about 1.3 % of women, as well as 2 % of men, are affected by PD in the USA, but the reason behind this male preponderance is still an inexplicit context (Ascherio and Schwarzschild, 2016).

PD is a multi-systemic disorder that is characterized by a combination of both motor as well as non-motor symptoms during its pathoprogression. The motor dysfunctions include consistent tremors, altered gait, slow movement (bradykinesia), disturbed balance, muscular rigidity, and postural instability (Poewe et al., 2017). Bradykinesia often leads to the appearance of a tendency for shuffling gait as well as the development of an expressionless face (hypomimia) accompanied by lower amplitudes of handwriting (micrographia) in PD (Sveinbjornsdottir, 2016). Apart from the motor impairment, several non-motor complications such as abnormal constipation, altered sleep-wake cycle, orthostatic hypotension, urinary disturbances, excessive hallucinations, loss of smell (anosmia), persistent sleeplessness (insomnia), cognition impairment, and especially dementia also occur in PD (Poewe et al., 2017; Marinus et al., 2018).

### *2.2. Etiology and risk factors of Parkinson's disease*

Although several theories and proposed mechanisms have been postulated regarding the possible etiological origin of PD, none has been conclusively proven so far. Among those theories, the relative contribution of genetic mechanisms, environmental factors, and lifestyle approaches remain conflicting (Poewe et al., 2017). On this etiological basis, PD has been classified into two broad categories, such as familial PD (having an inheritance in either the autosomal dominant or autosomal recessive pattern) and sporadic/idiopathic PD (having an origin from environment-gene interaction) (Ball et al., 2019). The familial PD constitutes about 15 % of total PD patients, who have an earlier family history, and 5–10 % of PD affected ones have a monogenic form of Mendelian inheritance. At least 23 genetic loci, in addition to 19 disease-causing genes, have been identified and further classified in PD till date (Deng et al., 2018). Major genes which are involved in the pathogenesis of PD include α-synuclein (SNCA/PARK1/4), Parkin (PRKN/PARK2), ubiquitin C-terminal hydrolase like 1 (UCHL1/PARK5), PTEN-induced kinase 1 (PINK1/ PARK6), DJ1 (PARK7), Leucine-rich repeat kinase 2 (LRRK2/PARK8), ATPase type 13A2 (ATP13A2/-PARK9), and GRB10 Interacting GYF Protein 2 (GIGYF2/ PARK11), *etc.* (Fig. 1) (Vázquez-Vélez and Zoghbi, 2021). Apart from familial PD, different environmental factors like prolonged use of pesticides, fertilizers, heavy metals, and several lifestyle diseases can cause the early onset of idiopathic/sporadic PD (Emamzadeh and Surguchov, 2018). For example, long-term exposure to pesticides like dieldrin (an organochlorine), rotenone (an organophosphate), and permethrin (synthetic pyrethroids) causes DA neuronal death and subsequent PD (Ball et al., 2019). In addition, a contaminant of synthetic heroin, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP), which is structurally similar to herbicide paraquat, is a potent neurotoxin to induce PD (Pang et al., 2019). Not only pesticides but also prolonged exposure to several heavy metals, including iron, copper, manganese, lead, and mercury, can potentially amplify the risk of PD (Ball et al., 2019). Apart from this, an increased risk of PD has often been correlated with a history of melanoma, the occurrence of type 2 diabetes mellitus, the incidence of traumatic brain injury, and consumption of dairy products (Ascherio and Schwarzschild, 2016). In contrast, a reduced PD risk has been reported in individuals consuming caffeine, smoking tobacco, and drinking alcohol (Pang et al., 2019).

### *2.3. Pathology and physiology of Parkinson's disease*

Although the exact mechanism for loss of DA neurons in SNpc is not well understood, several proposed theories support the subcellular impairment in DA neurons. The pathoclinical results demonstrate the Lewy bodies containing an aggregation of misfolded α-synuclein as the major hallmark of PD (Bloem et al., 2021). In addition, mitochondrial damage, proteasomal impairment, and autophagy dysfunction result from a complex interaction of several molecules, including α-synuclein

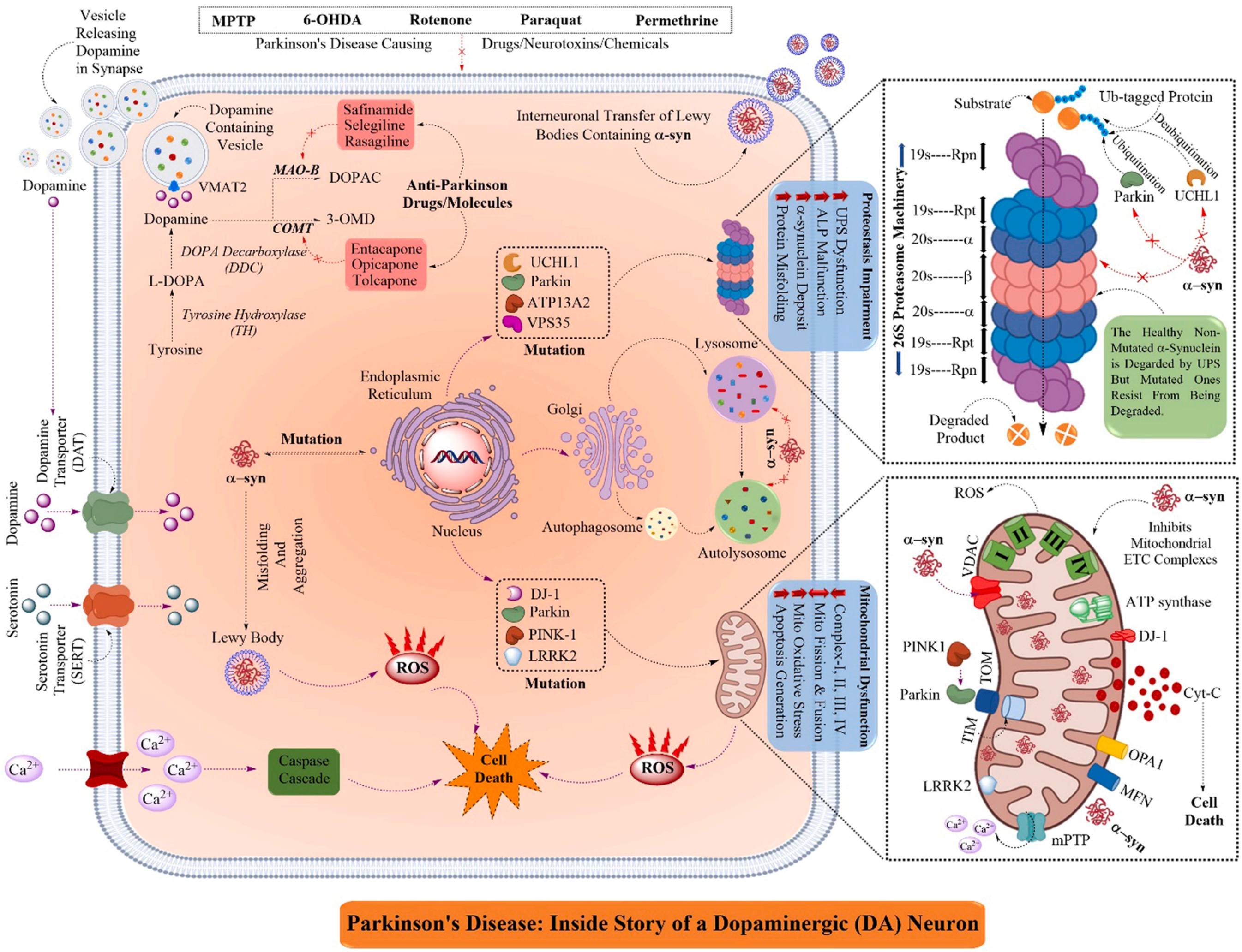


**Fig. 1. Schematic illustration of cellular & molecular mechanism of Parkinson's disease (PD):** PD is caused by the death of DA neurons in SNpc areas by the mutations and misfolding of several proteins, including α-synuclein. The mutations happen due to genetic reasons or environmental toxins, including MPTP, 6-OHDA, Rotenone, *etc.* The misfolding and aggregation of α-synuclein affect several subcellular organelles as well as systems, including mitochondria, proteasomes, and autophagosomes. The α-synuclein inhibits several components of proteasome, thus hindering its degradation and further accumulation of α-synuclein. Similarly, α-synuclein in mitochondria enters the organelle through VDAC, thus inhibiting ATP production and further potentiating ROS production and cytochrome C release, leading to apoptosis. In addition, α-synuclein also inhibits the fusion of lysosome with phagosome interfering with the intracellular clearance mechanism. [α-syn, Alpha-synuclein; PINK-1, PTEN-induced kinase 1; LRRK2, Leucine-rich repeat kinase 2; UCHL1, Ubiquitin C-terminal hydrolase L1; ATP13A2, ATPase cation transporting 13A2; VPS35, Vacuolar protein sorting ortholog 35; OPA1, Optic atrophy-1; MFN, Mitofusin; mPTP, Mitochondrial permeability transition pore; VDAC, Voltage-dependent anion channels; TIM, Translocase of the inner membrane; TOM, Translocase of the outer membrane, MAO-B, Monoamine oxidase B; COMT, Catechol-O-methyltransferase].

(Kalia and Lang, 2015). For example, α-synuclein is also observed to interact with several mitochondrial fusion proteins, including optic atrophy 1 (OPA1), mitofusin (MFN) 1, and 2, thus causing increased mitochondrial fission as well as elevated mitochondrial fragmentation. Moreover, α-synuclein can be translocated into the inner mitochondrial membrane (IMM) *via* voltage-dependent anion channels (VDACs) and subsequently inhibits their ion channel activity (Fig. 1) (Rocha et al., 2018). In addition, α-synuclein has also been reported to interact with various components of the outer mitochondrial membrane (OMM), such as translocase of the outer mitochondrial membrane (TOM) complex and the sorting and assembly machinery (SAM) complex, thus damaging the mitochondrial protein transport system. By damaging the transport system, α-synuclein can subsequently aggregate in intermembrane space (IMS), causing depolarization of the mitochondrial membrane and further generation of reactive oxygen species (ROS) (Pozo Devoto and Falzone, 2017). Most importantly, α-synuclein impairs the respiratory capacity as well as ATP production by inhibiting complex I activity along with inducing the mitochondrial permeability transition pore (mPTP) opening, subsequent cytochrome C release, and calcium influx, thus resulting in mitochondrial swelling and eventually neuronal apoptosis (Malpartida et al., 2021). It is evident that the mutated α-synuclein hinders the regular vesicular pathway, where it not only inhibits the docking of endoplasmic reticulum (ER) vesicles to the Golgi but also hinders the movement of vesicles from Golgi to lysosomes, causing ER stress (Vázquez-Vélez and Zoghbi, 2021). Mechanistically, α-synuclein interacts and inhibits the activating transcription factor 6 (ATF6), preventing its association with coat protein complex II (COPII). The inhibition of ATF6 can potentially hinder ER-associated protein degradation (ERAD), resulting in ER stress and, ultimately, neuronal apoptosis (Lehtonen et al., 2019). The clearance of α-synuclein remains an essential aspect of the pathoprogression of PD, where its cellular clearance is mediated by either ubiquitin proteasomal system (UPS) or autophagy lysosomal pathway (ALP). Surprisingly, the mutated α-synuclein not only resists proteasomal degradation by damaging several UPS components but also impairs autophagy, thus augmenting the further accumulation of this protein inside the neurons (Fig. 1.) (Olanow and McNaught, 2006).

## 3. Blood platelet: a peripheral model for Parkinson's disease (PD)

Growing evidence in the field of neurobiology is considering a robust association between NDDs and CVDs. Among other NDDs, PD has several non-genetic risk factors, including lifestyle diseases, where the reports of cardiovascular complications are establishing a novel field of interest (Firoz et al., 2015). The underlying pathoprogression associated with nearly all CVDs has an origin from atherosclerosis, which results in the development diseases like cardiac arrhythmias, coronary artery disease, myocardial infarction, and ischemic stroke (Flora and Nayak, 2019). Atherosclerosis and associated etiological cellular factors bring platelet as a highlighting blood component for the pathoprogression of PD that is further discussed in the below sections.

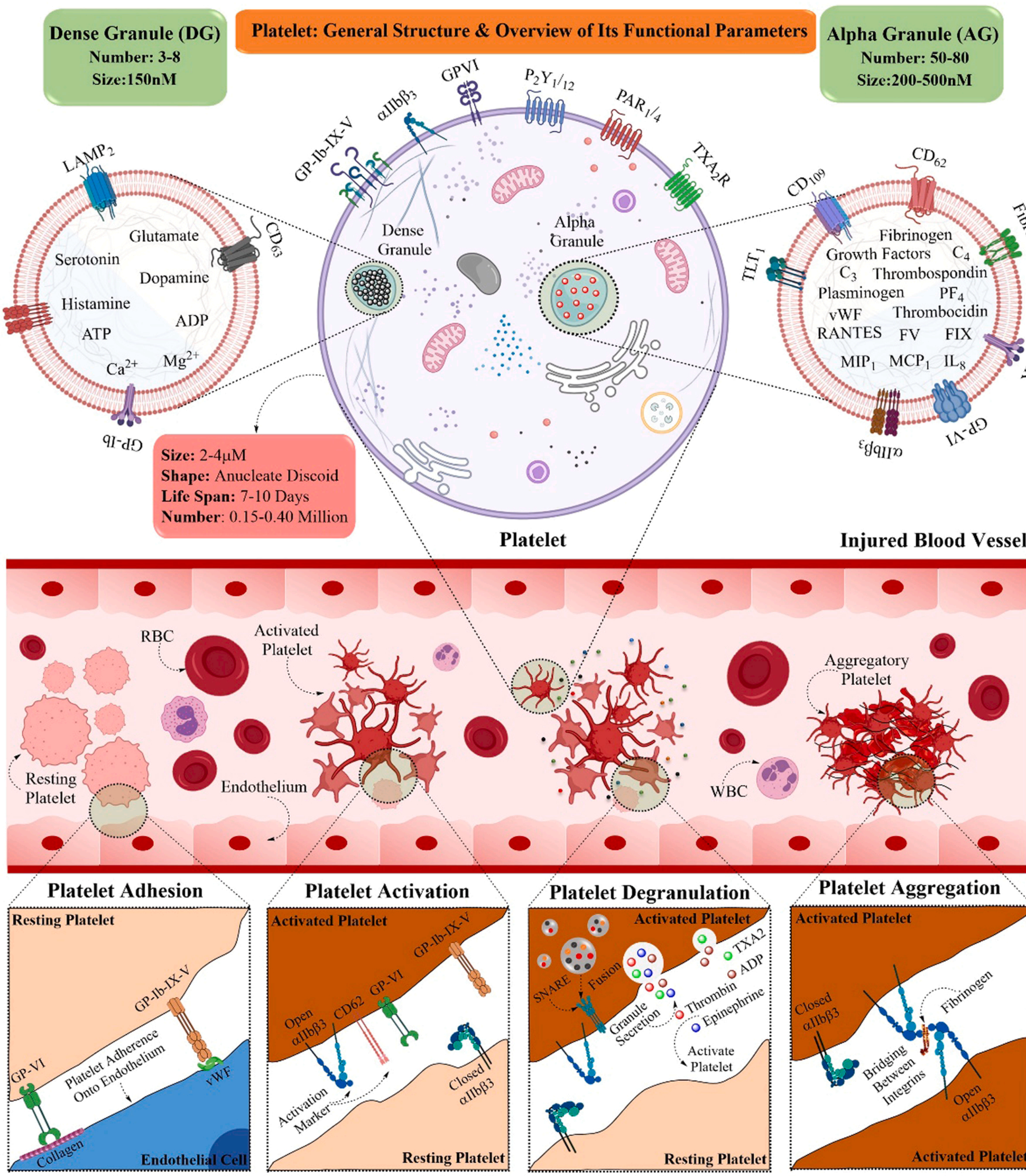


**Fig. 2. Schematic representation of granular contents of platelet and platelet functional parameters:** Platelet is a tiny blood component that helps in hemostasis and thrombosis. After any vascular injury, the platelet adheres to the injured endothelium *via* its surface glycoproteins. Following adhesion, platelets become activated and secrete several agonists like TXA2, Thrombin, ADP, *etc.* These agonists further activate other platelets, eventually aggregating to each other by integrin αIIbβ3 and fibrinogen binding. Platelets possess two prominent granules, *i.e.,* alpha- (carrying protein molecules) and dense-granules (carrying non-protein molecules), which help in achieving various major platelet functions. [LAMP2, Lysosome-associated membrane protein 2; vWF, von Willebrand factor; TXA2, Thromboxane A2; RANTES, Regulated upon activation, normal T cell expressed and presumably secreted; FV, Factor V; FIX, Factor IX; IL-8, Interleukin 8; MCP-1, Monocyte chemoattractant protein-1; MIP-1, Macrophage inflammatory protein-1; PAR1/12, Protease-activated receptors 1/12; TXA2R, Thromboxane A2 receptor; TLT-1, TREM like transcript-1].

### 3.1. Association of vascular complications in Parkinson's disease

Vascular complications, especially CVDs, have a significant impact on the onset of PD. Though brain and heart tissues vary a lot in their structures and functions, but their excitability, energy metabolism, and cellular division bring them to the same table (Firoz et al., 2015). Endothelial dysfunction (ED) is regarded as the initial event that happens after a vascular injury and further triggers several cellular and molecular cascades to form an atherosclerotic plaque (Tana et al., 2018). As the atherosclerotic plaques start rupturing during ED, these plaques, along with the coagulation system, initiate the formation of a fibrin clot, called a thrombus, where a blood cell fragment known as platelet plays a predominant role to prevent excessive blood loss (Koupenova et al., 2017). Furthermore, persistent ED accompanied by elevated blood levels of proinflammatory molecules such as IL-1β, IL-6, and TNF-α could be a predominant thrombotic risk in PD patients (Tana et al., 2018). Several classical determinants for CVDs are also considered the potential risk factors for PD, including hypertension, obesity, hypercholesterolemia, and diabetes mellitus (Potashkin et al., 2020). Nevertheless, clinical PD patients are reported to have a higher risk of CVDs, including myocardial infarction, ischemic stroke, coronary heart disease, and congestive heart failure (Park et al., 2020). Therefore, the blood platelet is indispensable for the development of cerebrovascular complications that can cause the early onset of PD. Platelets are tiny non-nucleated cellular fragments circulating in the blood that primarily contributes to hemostasis, *i.e.*, the process of stopping bleeding at the site of injured endothelium. In order to accomplish this event, platelet undergoes four major but coinciding processes of different signaling mechanisms such as adhesion, activation, secretion, and aggregation (Fig. 2) (Li et al., 2010; van der Meijden and Heemskerk, 2019). Apart from hemostasis, platelet also plays a pivotal role in regulating immune functions such as inflammation and phagocytosis (Morrell et al., 2014). But the exact mechanism of association between CVDs and PD is still elusive and the role of platelet sometimes remains controversial or even counter-intuitive.

### 3.2. Blood platelet as a peripheral model for Parkinson's disease

Platelets are nowadays gaining remarkable recognition for their role beyond hemostasis after vascular injury, thus signifying their pivotal contribution in providing intercellular communication between the brain and blood (Leiter and Walker, 2020). Emerging research in the field of platelet biology has demonstrated a striking resemblance between platelet and neuron both in their pattern of degranulation or secretion, vesicular trafficking, as well as expression of surface receptors (Beura et al., 2022). Surprisingly, the alpha-granules of platelets are comparable to that of large dense-core synaptic vesicles of neurons. In contrast, the dense-granules of platelets resemble that of small dense-core synaptic vesicles of neurons (Leiter and Walker, 2020). Furthermore, the platelets also store and secrete various neurotransmitters, including glutamate, serotonin, dopamine, histamine, epinephrine, γ-aminobutyric acid (GABA), as well as ADP, ATP, and $Ca^{2+}$ in their dense-granules (Ponomarev, 2018). Not only the neurotransmitters but also the receptors as well as transporters for dopamine (vesicular monoamine transporter 2/VMAT2), glutamate (excitatory amino acid transporters/EAAT), serotonin (serotonin transporter/SERT), GABA (GABA-betaine transporter 1/BGT1), and choline (choline transporter-like protein 1/CTL1) are also present in blood platelets (Jedlitschky et al., 2012). In addition, platelets also express, store and secrete several neuron-specific proteins such as amyloid precursor protein (APP), amyloid-beta (Aβ), tau, brain-derived neurotrophic factor (BDNF), monoamine oxidase (MAO), reelin, and ephrin (Beura et al., 2022). In a study, it has been confirmed that platelet is the primary source of circulatory Aβ peptide in human blood, where about 90 % of total circulatory Aβ comes from or is stored in platelets (Inyushin et al., 2020). Apart from this, platelet has been gaining recognition as a successful model system for studying pathoprogression of PD owing to its storage of a major PD-biomarker, α-synuclein (Fig. 4). In fact, the highest concentration of α-synuclein per milligram of cellular protein in blood is found in these small platelets (Tashkandi et al., 2018; Stefaniuk et al., 2021). Moreover, various PD-specific biomarkers, including tyrosine hydroxylase (TH) (Drozdov and Anokhina, 1990), dopamine transporter (DAT) (Frankhauser et al., 2006), vacuolar protein sorting ortholog 35 (VPS35) (Rijkers et al., 2017), lysosomal associated membrane protein-2 (LAMP-2) (Israels et al., 1996), parkin (PRKN) (Lee et al., 2020), PTEN-induced kinase 1 (PINK1) (Walsh et al., 2018), protein deglycase DJ-1 (Shi et al., 2010) are reported to be expressed by platelet, thus making it a suitable peripheral model for PD (Fig. 3). Therefore, the study of platelet remains fascinating in the pathoprogression of PD.

## 4. Role of blood platelet in Parkinson's disease (PD)

Keeping in account the above biomarkers, the versatile role of platelet, highlighting its apparent systemic involvement in PD, has been markedly recognized. As a result, it becomes indispensable to focus on the significance of platelet morphology as well as physiology in PD. It can thus be understood that platelets could be our potential allies in the near future, by elucidating the effect of its different structural and functional alterations in PD (Behari and Shrivastava, 2013). Besides this, platelet secretory components also modulate the disease pathoprogression, which is discussed in the following sections (Fig. 3).

### 4.1. Platelet structural, functional and subcellular organelle impairments in PD

It is evident now that a platelet shares significant similarities with a neuron, but in addition to that, the platelet also functions as a peripheral model for AD. In AD, several structural, as well as functional alterations of platelets, occur (Table 1) (Gowert et al., 2014). Like AD, PD has also been associated with several structural as well as functional changes in platelet. The platelet count is an important clinical parameter, which decides the severity of an associated disease, and a decreased platelet number (thrombocytopenia) has been reported in PD patients (Machaczka et al., 1999; Koçer et al., 2013; Behari and Shrivastava, 2013). In contrast to the previous findings, Sharma et al. have reported that platelet count remains unchanged in PD patients (Sharma et al., 1991). During PD, the platelet swells and increases in size to become larger (Pei and Maitta, 2019; Simonidze and Samushia, 2014), which further gets confirmed, when α-synuclein-deficient mice reported small-sized platelets accompanied by extensive fragmentation and degranulation of those platelets (Pei and Maitta, 2019). Apart from an increase in size, the platelet shape is also observed to be distorted and stretched, having less membrane invagination in PD patients, thus validating platelet abnormal structural changes in PD (Simonidze and Samushia, 2014). In addition, the larger platelet size is caused by an increased cytoplasmic volume, which is observed in PD patients (Koçer et al., 2013; Ferrer-Raventós and Beyer, 2021). Along with an increased platelet volume, the platelet vacuoles of PD patients are also confirmed to be abnormally bigger than those of healthy individuals, suggesting a possible mechanism for increased platelet volume (Factor et al., 1994). The larger platelet size is also confirmed *via* the measurement of mean platelet volume (MPV), which is an index of average platelet size. An increased MPV is reported in PD patients as compared to healthy individuals, which was further supported by the finding of elevated MPV in the presence of exogenous α-synuclein (Koçer et al., 2013; Geyik et al., 2016; Tashkandi et al., 2018). Apart from the structural impairments, platelet functional alterations are also observed in PD patients, including changes in adhesion, activation, aggregation, and degranulation of platelet. Whenever an injury occurs, the circulating platelets attach to extracellular matrix substances outside the injured endothelium *via* a distinct process called adhesion. For example, platelet membrane

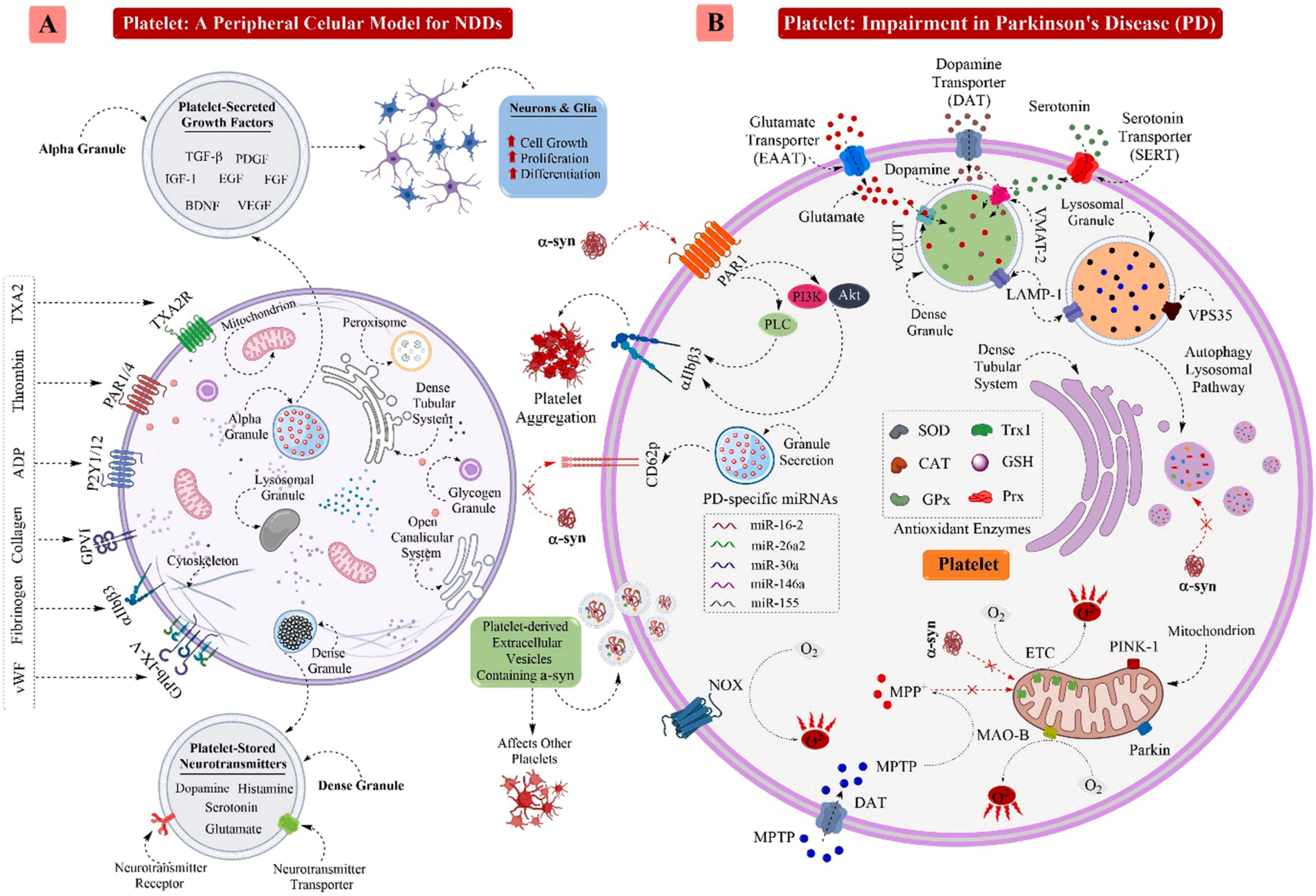


**Fig. 3. Schematic depiction about major alterations in platelet during Parkinson's disease (PD):** A generalized structure of human platelet having several key surface receptors with their binding agonists, different granules such as dense-granule and alpha-granule as well as membranous structures like dense tubular system, open canalicular system, and mitochondria **(A).** The α-synuclein targets several components of a platelet, including the inhibition of PAR1 mediated platelet aggregation, mitochondrial ETC complex inhibition, and increased ROS production. Several neurotoxic compounds such as MPTP enters platelet *via* DAT and further inhibits mitochondrial complexes. Alteration in uptake and storage of dopamine, serotonin, and glutamate also occurs in PD platelet. Platelet carries several antioxidant enzymes, PD-specific microRNAs, and extracellular vesicles containing α-synuclein. Platelet-rich growth factors from alpha-granules help in neurogenesis in PD, and various antiplatelet drugs are also proven effective against PD **(B).** [α-syn, Alpha-synuclein; vWF, von Willebrand factor; TXA2, Thromboxane A2; PAR1, Protease-activated receptor 1; TXA2R, Thromboxane A2 receptor; NOX, NAD(P)H oxidase; GPx, Glutathione peroxidase; SOD, Superoxide dismutase; XOD, Xanthine oxidase; CAT, Catalase; GSH, Glutathione; Prx, Peroxiredoxin; Trx1, Thioredoxin 1; MAO-B, Monoamine oxidase B; PINK-1, PTEN-induced kinase 1; LAMP1, Lysosome-associated membrane protein 2; vGLUT, Vesicular glutamate transporter; VMAT2, Vesicular monoamine transporter 2; ETC, Electron transport chain; VPS35, Vacuolar protein sorting ortholog 35; PDGF, Platelet-derived growth factor; IGF-1, Insulin-like growth factor 1; FGF, Fibroblast growth factor; BDNF, Brain-derived neurotrophic factor; VEGF, Vascular endothelial growth factor; TGF-β, Transforming growth factor-β].

receptor-like GPIb-IX-V binds with vWF, and integrin α2β1, GPVI, and GPIV bind with collagen in addition to integrin α5β1 and αIIbβ3, which bind with fibronectin of the exposed extracellular matrix components for platelet adhesion (Ruggeri and Mendolicchio, 2007). In a study, Reheman et al. reported a defective platelet adhesion at the site of vascular injury in α-synuclein deficient mice, thus forming defective platelet-rich thrombi (Reheman et al., 2010). A defective platelet adhesion often results in altered platelet activation through several mechanisms. In a recent study, platelet hyperactivation was demonstrated to significantly contribute to the hypercoagulability of blood platelets in PD patients as compared to healthy control (Adams et al., 2019). Furthermore, platelet hypercoagulability indicates the tendency of blood platelets to develop not only thrombus formation but also thrombus-associated complications in PD (Tashkandi et al., 2018). Nevertheless, the platelet activation is manifested by several soluble circulating agonists like ADP, thromboxane $A_2$ ($TXA_2$), and thrombin. For example, the ADP released from damaged endothelial cells activates platelets *via* platelet receptors $P2Y_1$ and $P2Y_{12}$, whereas $TXA_2$ regulates platelet activation through binding with platelet $TXA_2$ receptor, TP, and thrombin *via* protease-activated receptor (PAR) on the platelet surface (Estevez and Du, 2017). Contrary to the previous reports, the exogenous treatment of α-synuclein potentially inhibited the thrombin- or ionomycin-induced expression of platelet surface P-selectin (CD62P), a platelet activation marker on purified platelets (Park et al., 2002). Besides activation, platelet aggregation is an important parameter in the disease perspective deciding the fate of a thrombus formation. Platelet aggregation is the process where one platelet binds with other platelets, which is predominantly mediated by bridging between platelet surface receptor, integrin αIIbβ3, *via* the dimeric adhesive protein fibrinogen (Jackson, 2007). Thus, platelets form plugs by aggregating to each other and are further associated with activation of the different coagulation cascades resulting in fibrin deposition and cross-linking (van der Meijden and Heemskerk, 2019). It has been reported that ADP- and epinephrine-induced platelet aggregation remains significantly decreased in PD patients (Sharma et al., 1991). When the above clinical finding was simulated *in vitro*, Pontarollo et al. found that exogenous human α-synuclein can also potentially inhibit platelet aggregation by interfering with the thrombin/PAR1 functional axis (Fig. 3) (Pontarollo et al., 2021). A similar study by Lim et al. have also reported that MPP+, a PD-inducing neurotoxin, can trigger the depletion of ATP, thus triggering decreased platelet aggregation in platelet-rich plasma (Lim et al., 2009). Further, platelets also contract their cytoskeleton to shrink the thrombus called a clot, thus making the completion of hemostasis (van der Meijden and Heemskerk, 2019). In a recent report, the platelet-mediated clot formation study has revealed that an increased clot rate together with an elevated platelet-regulated fibrin cross-linking as well as a reduced time of thrombus generation generally occurs in PD patients (Adams et al., 2019). In addition, PD patients are associated with a decreased secretion of α-granular contents because of a low amount of granules in platelet (Simonidze and Samushia, 2014). Moreover, the decreased degranulation is further confirmed by Park et al., and the same group demonstrated that exogenous α-synuclein could significantly inhibit the $Ca^{2+}$-dependent release of alpha-granules in purified platelets. Still, the release of dense-granules as well as lysosomal-granules remains unaffected by α-synuclein (Park et al., 2002). The platelet plasma membrane plays an important role in platelet secretion for various physiological functions. In a study, it has been found that exogenous α-synuclein can enter into platelets and localizes itself majorly near the plasma membrane and cytosol like the same in neurons (Hashimoto et al., 1997; Park et al., 2002). Unlike AD, the membrane fluidity of PD platelets remains unchanged, suggesting the balanced integrity of the platelet membrane during PD pathoprogression (Factor et al., 1994). In an ultrastructure study, Factor et al. reported larger platelet vacuoles, which are the part of open canalicular system (OCS) in PD platelets, suggesting an alteration of vesicle trafficking in platelet (Factor et al., 1994). OCS is an invaginated form of platelet-delimited plasma membrane that helps in intraplatelet cargo transport and vesicular trafficking (Twomey et al., 2018). In addition to this, mitochondrial dysfunction and reduced energy production are the evident phenomena in PD, where the reduced production of ATP due to decreased activities of mitochondrial complexes is reported (Poewe et al., 2017). Similarly, a reduced maximal electron transport system (ETS) capacity as well as compromised respiration rates in platelets is observed due to decreased activity of mitochondrial complex I, II, and III in platelets of PD patients (Fig. 3) (Haas et al., 1995; Ferrer-Raventós and Beyer, 2021). Furthermore, a decreased activity of coenzyme Q10 (ubiquinone) is also reported in PD platelets accompanied by an increased proportion of oxidized coenzyme Q10 (Behari and Shrivastava, 2013; Clark et al., 2021). All these platelet-associated changes signify the impairment of platelet along with its organelles in PD patients.

### 4.2. *Platelet storage contents and secretory components in pathogenesis of PD*

Being a prominent secretory cell, platelet releases various bioactive substances from different storage granules such as alpha-granules, dense-granules, and lysosomal-granules to mediate a wide array of pathophysiological processes apart from hemostasis, like inflammation, angiogenesis, and tissue repair in several diseases (Beura et al., 2022). The dense-granules are comprised of amine molecules such as serotonin, histamine, dopamine, and purines like ATP and ADP, as well as cations including $Mg^{2+}$, $Ca^{2+}$, *etc.* Similar to dense-granules, the alpha-granules possess a wide range of protein molecules like growth factors, coagulation molecules, inflammatory molecules along with hemostatic factors (Yadav and Storrie, 2017; Heijnen and Van der Sluijs, 2015). In addition, platelet can uptake different cellular as well as plasma proteins from the blood circulation, thus providing an alternative mechanistic insight into disease pathogenesis (van der Meijden and Heemskerk, 2019). For example, platelet contains the highest concentration of α-synuclein per milligram of cellular protein in the blood. However, most of these α-synuclein proteins are found in RBCs, being the most abundant cells in the blood (Tashkandi et al., 2018; Pei and Maitta, 2019). The α-synuclein in platelets is reported to be localized into alpha-granules and is also present in platelet-derived extracellular vesicles, where it regulates platelet vesicle transfer and granule secretion (Pienimaeki-Roemer et al., 2015). Furthermore, the α-synuclein structure in platelet is reported to be similar to those found in neurons having an N-terminus, central region, and C-terminus end (Stefaniuk et al., 2021). Apart from α-synuclein, the human platelets also contain several PD-specific neurotransmitters and their receptors. For example, platelets express dopamine, dopamine receptors (Beura et al., 2022), as well as dopamine transporters (DAT) identical to that of neuronal origins, which are associated with the regulation of platelet behavior (Frankhauser et al., 2006). Moreover, nearly 99 % of the circulating dopamine is concentrated inside blood platelets in their cytoplasm *via* DAT. After transportation into the cytoplasm, the dopamine is further taken up into the dense-granules for storage until it is secreted upon platelet activation (Fig. 3) (Osinga et al., 2016). As PD is associated with reduced dopamine levels in the brain, a reduced dopamine uptake has also been reported by the storage granules of platelets, which is speculated to be regulated by DAT in PD patients (Rabey et al., 1993). The dopamine transport into dense-granules of platelets is mediated by vesicular monoamine transporter 2 (VMAT2), and surprisingly, the VMAT2 mRNA level is reported to be decreased in platelets of PD patients (Sala et al., 2010; Osinga et al., 2016). Besides dopamine, dopamine transporters and dopamine receptors, platelet also possesses tyrosine hydroxylase (TH), which is the rate-limiting enzyme for the biogenesis of dopamine from tyrosine (Drozdov and Anokhina, 1990). In addition to dopamine, glutamate is an important neurotransmitter for memory establishment and dementia, which is also associated with PD. Platelets express not only glutamate in their dense-granules but also the glutamate receptors on their surface (Gautam et al., 2019). It has been reported that an increased glutamate level is found in platelets of PD patients, but surprisingly the glutamate reuptake is confirmed to be decreased in those PD platelets (Ferrarese et al., 1999). This finding suggests that platelet might be synthesizing more amount of glutamate in PD patients on its own, as it is not taking glutamate from blood plasma. In a study, Bonuccelli et al. demonstrated that PD is associated with a reduced expression of benzodiazepine receptors, a part of the GABA receptor complex present on the outer membrane of mitochondria (Bonuccelli et al., 1991). Apart from α-synuclein, there are various other PD-specific proteins, which are stored and secreted from platelets. For example, in a recent study, Lee et al. have reported that parkin (PRKN/PARK2), an E3 ubiquitin ligase is highly expressed in platelets compared to other tissues, where it helps in autophagy, especially mitophagy (Lee et al., 2020). In addition to Parkin, another PD-specific protein, PINK1 (PARK6), a mitochondrial enzyme, can potentially regulate platelet activation, where its deficiency could significantly increase platelet activation and thrombosis, thus supporting the earlier finding of increased activation of blood platelets in PD (Walsh et al., 2018). PINK1 is a serine/threonine-protein kinase found in mitochondria, which helps in the quality control of mitochondrial activity by regulating the degradation of damaged mitochondria *via* mitophagy (Pickrell and Youle, 2015). It is also reported that the PINK1-triggered mitophagy can significantly reduce platelet oxidative stress (Walsh et al., 2018). Moreover platelets are also confirmed to store and express an essential PD-specific protein deglycase DJ1 (PARK7), which functions as an intracellular redox-sensitive chaperone as well as an oxidative stress sensor (Shi et al., 2010; Goelz et al., 2021). Furthermore, platelets also express an activation-dependent surface glycoprotein, LAMP-2, on their lysosomal-granules as well as dense-granules for degradation of substrates *via* the lysosomes, in a process known as chaperone-mediated autophagy (CMA) (Silverstein and Febbraio, 1992; Israels et al., 1996). Platelets also have a PD-specific protein, VPS35, on their lysosomal-granules, which helps in vesicular cargo transport and autophagy (Rijkers et al., 2017). In a study, Babur et al. searched for the detection of either PD-specific LRRK1 or LRRK2 in platelets but could not find the same proteins in platelet (Babur et al., 2020). LRRK is a cytoplasmic kinase with multiple functional domains, where it interacts with several cytosolic proteins to regulate vesicle endocytosis, cytoskeleton maintenance, neurotransmitter release, and their uptake (Klein and Westenberger, 2012). Platelets, like neurons, carry monoamine oxidase-B (MAO-B), a flavin oxidase found on the outer mitochondrial membrane that inhibits the arylalkylamine-dependent release of serotonin in platelets, thus enhancing its storage (Bonuccelli et al., 1990). Moreover, MAO-B is more likely to generate ROS as a by-product *via* oxidative deamination of several neuroactive amines, including dopamine. In PD, the platelet MAO-B activity was reported to be significantly higher (Danielczyk et al., 1988; Zhou et al., 2001), suggesting a theory of decreased dopamine in platelet as well as increased ROS, and elevated oxidative stress along with subsequent ROS-mediated platelet hyperactivity. The platelets are also equipped with several key antioxidant enzymes in order to counter the ROS generation, including glutathione peroxidase (GPx), superoxide dismutase (SOD), catalase (CAT), peroxiredoxin (Prx), glutathione (GSH), thioredoxin 1 (Trx1), and thioredoxin reductase (TrxR) (Fig. 3) (Pietraforte et al., 2014; Masselli et al., 2020). A reduced activity of SOD (Behari and Shrivastava, 2013), and GPx (De La Fourniere et al., 2000) is observed in PD platelets, hereby explaining a possibility of increased ROS in platelets like as neurons in the brain. Further experiments will definitely elucidate more about the antioxidant status of blood platelets in PD condition. Hence, it is obvious that platelet expresses as well as stores several key secretory components, which are associated with PD, and an altered granular secretion of platelet often occurs during PD.

## 5. Theranostic potential of platelet in Parkinson's disease (PD)

PD is regarded as a complex disease with a broad spectrum of motor and non-motor symptoms, which require individualized therapeutic approaches. As each PD patient varies in his/her symptoms, there exists no standard treatment for PD to date. Even after possible treatments, PD patients report having debilitating response fluctuations, which require efficient device-aided therapies aimed to stimulate DA neurons like neurosurgical interventions targeting the areas of basal ganglia, with either high-frequency stimulation called as deep brain stimulation and/or precise lesion surgery (Bloem et al., 2021). The available medications and probable surgery can only relieve the symptoms and maintain a better quality of life. This is because of the challenges in early diagnosis, as the early motor symptoms in PD appear after the loss of almost 50–60 % of DA neurons in the substantia nigra accompanied by deprivation of 70 % dopamine in the striatum (Ugrumov, 2020).

### *5.1. Blood platelet as a peripheral diagnostic model for Parkinson's disease*

Although PD has no definitive test for a clear diagnosis, it is partially diagnosed on the basis of clinical criteria. In clinical practice, the strategy for diagnosis includes the presence of multiple cardinal motor symptoms along with associated and exclusionary symptoms and response to levodopa (L-DOPA) (Bloem et al., 2021). The gold standard for PD diagnosis is the neuropathological assessment that utilizes neuroimaging techniques to decipher DA neuronal death and dopamine deficiency. The potential candidate imaging markers include positron emission tomography (PET) or single-photon emission computed tomography (SPECT) methods to evaluate the decreased projection of SNpc-DA nerve terminals to the striatum (Kalia and Lang, 2015). However, studies have assessed several clinical, biochemical, and genetic biomarkers in cerebrospinal fluid (CSF) as well as blood for the early detection of PD. For example, the differential levels of α-synuclein in the blood can be a useful breakthrough for risk prediction in screening and diagnosis for measuring the progression of PD and evaluating its potential therapeutic regimes (Poewe et al., 2017). Therefore, platelet as a circulatory blood component can be used as a peripheral cellular model for the early diagnosis of PD like that of AD earlier. In AD, platelet has been proposed to store and secrete several key AD-specific molecules, which can be used not only as diagnostic molecules but also as therapeutic options (Veitinger et al., 2014). However, platelet stores and secretes the predominant clinical PD biomarker, α-synuclein, suggesting the potential value of platelet to be used for early disease assessment (Tashkandi et al., 2018). A recent finding suggests that α-synuclein in extracellular vesicles of blood plasma may serve as an early diagnostic biomarker for PD (Stuendl et al., 2021). Surprisingly, about 70–90 % of all circulatory extracellular vesicles are secreted by platelets (Teixeira et al., 2016). These α-synucleins are also reported to be present in platelet-derived extracellular microvesicles (Table 2) (Pei and Maitta, 2019). PD has been associated with a diminished level of dopamine and serotonin in the striatum, which is also decreased in peripheral circulatory platelets, thus making it a suitable diagnostic model for PD (Stahl, 1977). Apart from dopamine, platelet contains several key PD-specific miRNAs, including miR-22-3p, miR-146a, and miR-155, and the level of these miRNAs are reported to be elevated in PD platelets (Espinosa-Parrilla et al., 2019). Another study has confirmed that a reduced expression of PD-specific miRNAs like miR-128-3p and miR-139-5p are significantly found in PD platelets (Gámez-Valero et al., 2021). Furthermore, several PD-related genes are also identified in peripheral blood platelets, including WASP Family Homolog 5 Pseudogene (*WASH5P*), Metastasis-associated Lung Adenocarcinoma Transcript 1 (*MALAT1*), Eukaryotic Elongation Factor 1A (*EEF1A1*), and Cathepsin S (*CTSS*), which can be the potential biomarkers for PD diagnosis (Zhang et al., 2021).

### *5.2. Platelet as a therapeutic solution for the treatment of Parkinson's disease*

There are several medications available to treat PD, although none can reverse the severity of the illness. These medications can only work to enhance dopamine synthesis, mimic the function of dopamine, or inhibit dopamine degradation in order to alleviate PD symptoms (Bloem et al., 2021). Therefore, the replenishment of striatal dopamine deficiency *via* the systemic administration of L-DOPA, the dopamine precursor amino acid, represents a gold standard treatment for PD (Poewe et al., 2017). But the peripheral administration of L-DOPA can be compromised by plasma catechol-O-methyltransferase (COMT) and DOPA decarboxylase (DDC) enzymes, which are responsible for L-DOPA metabolism in the circulation before entering into the brain. Hence, L-DOPA, in combination with COMT inhibitors (Entacapone, Opicapone, and Tolcapone) and DDC inhibitor (Carbidopa), is approved by the Food and Drug Administration (FDA) for the treatment of PD (Kalia and Lang, 2015), (Emamzadeh and Surguchov, 2018). Apart from these, the other FDA approved therapeutic treatment for PD includes dopamine agonists (*e.g.*, Pramipexole, Ropinirole, Rotigotine, and Apomorphine) to mimic the effect of dopamine in DA neurons, MAO-B inhibitors (Selegiline, Rasagiline, and Safinamide) to reduce dopamine degradation and ROS formation (Armstrong and Okun, 2020; Poewe et al., 2017) serotonin 5-HT2A receptor antagonist (Pimavanserin) to reduce hallucinations and delusions, and adenosine A2A receptor antagonist (Istradefylline) (Emamzadeh and Surguchov, 2018; Sivanandy et al., 2021) to enhance the secretion of GABA (Fig. 4). Besides these well-established drugs, new add-on medications can be helpful to counter various PD-associated complications and risk factors such as CVDs. Among such, antiplatelet drugs can be the possible alternate options, which could be used along with the frontline drugs for better disease management.

#### *5.2.1. Potential antiplatelet drugs as therapeutic options for Parkinson's disease*

There are numerous antiplatelet drugs that are used in several CVDs intended to reduce the risk of thrombosis and associated complications. These drugs affect several platelet signaling cascades and inhibit platelet activation as well as aggregation (Fig. 4) (Michelson, 2010). Few such drugs are described below, having a probable efficacy in PD by lowering neurotoxicity and promoting neurogenesis in both *in vitro* as well as *in vivo* PD models (Table 2).

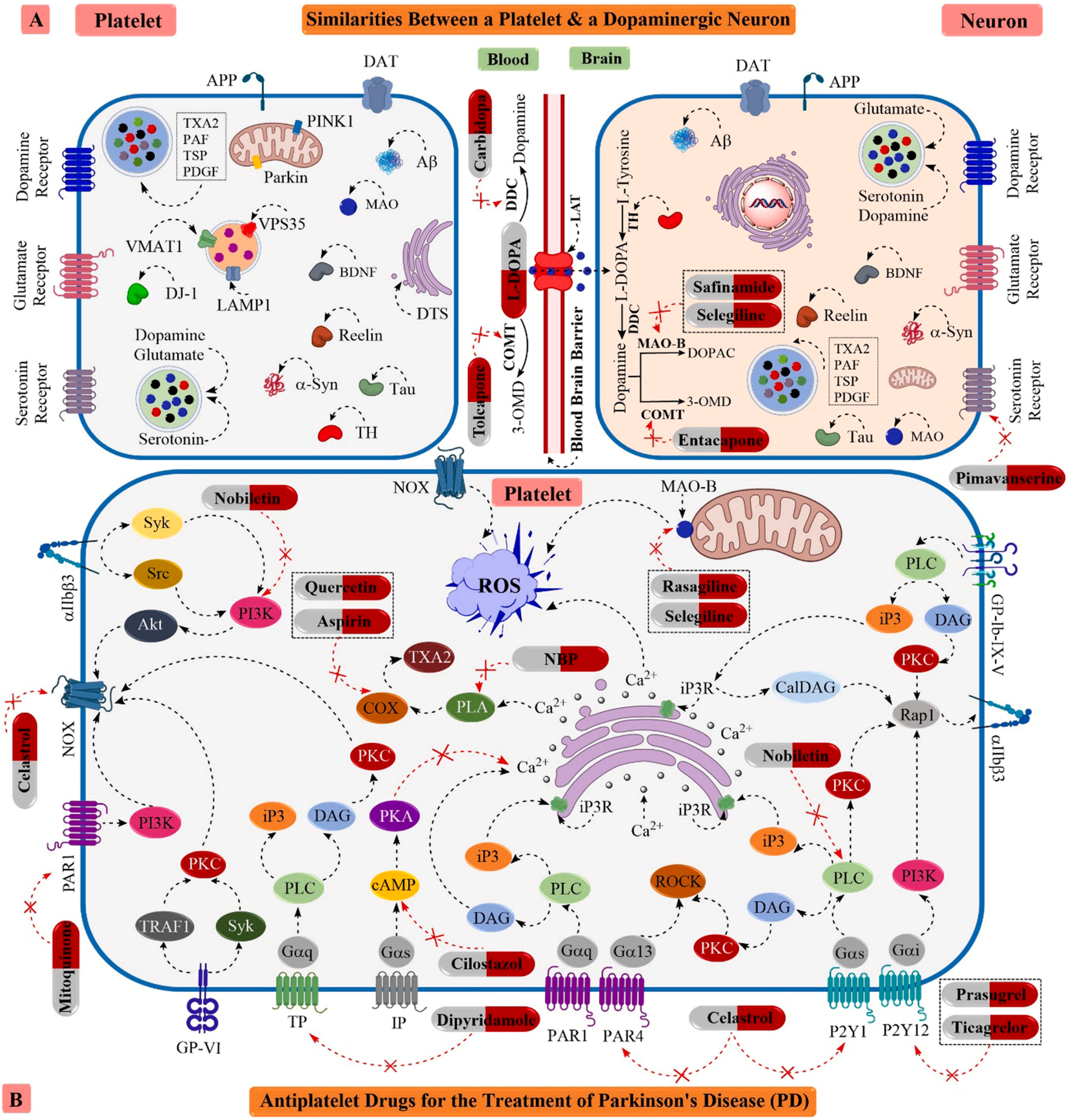


**Fig. 4. Schematic overview of the platelet-associated theranostic solutions for the management of Parkinson's disease (PD):** Platelet is considered a peripheral cellular model for PD, as it shares significant similarities with a dopaminergic (DA) neuron. For example, platelet contains α-synuclein, parkin, DJ-1, PINK-1, BDNF, Reelin, VPS35, LAMP1, VMAT-1, dopamine, serotonin, DAT, TH, *etc.* These striking resemblance makes platelet a suitable surrogate system to study more about the mechanism of PD pathoprogression and the development of novel therapeutic candidates for PD. Several FDA-approved drugs have been used to restore dopamine levels inside a DA neuron for the management of PD, including L-DOPA, Carbidopa, Tolcapone, Safinamide, Selegiline, *etc.* **(A).** Antiplatelet drugs have been used for a long time to prevent thrombus formation by inhibiting platelet activation and aggregation, targeting several key platelet agonist receptors as well as downstream signaling pathways. Few drugs like Celastrol, Aspirin, Prasugrel, Ticagrelor, Cilostazol, Dipyridamole, Mitoquinone, *etc.*, are shown with their mechanism of action to inhibit platelet activation and subsequent thrombus formation **(B).** [MAO-B, Monoamine oxidase B; PINK-1, PTEN-induced kinase 1; LAMP1, Lysosome-associated membrane protein 2; vGLUT, Vesicular glutamate transporter; VMAT2, Vesicular monoamine transporter 2; VPS35, Vacuolar protein sorting ortholog 35; PAR1/12, Protease-activated receptors 1/12; APP, Amyloid precursor protein; DAT, Dopamine transporter; BDNF, Brain-derived neurotrophic factor; LAT, L-amino acid transporter; TXA2, Thromboxane A2; TP, TXA2 receptor; IP, PGI2 receptor; PAF, Platelet-activating factor; TSP, Thrombospondin; PDGF, Platelet-derived growth factor; DDC, DOPA decarboxylase; COMT, Catechol-O-methyltransferase; TH, Tyrosine hydroxylase; NOX, NAD(P)H oxidase; DOPAC, 3, 4-Dihydroxyphenylacetic acid; 3-OMD, 3-O-methyldopa; NBP, 3-*N*-butylphthalide].

**Table 1**

**Structural, functional & secretory alteration of platelet in Parkinson's disease:** A detailed description of alterations of platelet structural and functional parameters accompanied by platelet organelle dysfunction as well as dysregulated secretion of PD specific proteins from platelet.

| Category | Parameters/Factors | Contributions in PD | Refs. |
|---|---|---|---|
| Platelet structural alterations | Platelet count | PD has been correlated with a change in platelet count. Plate count was reported to be decreased in the case of PD patients.<br>PD patients are not associated with a low platelet count. | (Machaczka et al., 1999; Behari and Shrivastava, 2013; Koçer et al., 2013)<br>(Sharma et al., 1991) |
| | Platelet size | PD is being correlated with an abnormal platelet size, where a larger size is observed in PD patients and a smaller size in α-synuclein deficient mice. | (Simonidze and Samushia, 2014; Tashkandi et al., 2018; Pei and Maitta, 2019) |
| | Platelet shape | PD is associated with an altered platelet shape. More membrane extension is observed in PD platelets. A minimum invagination and distorted shape are observed in PD patients. | (Simonidze and Samushia, 2014; Pei and Maitta, 2019; Adams et al., 2019) |
| | Platelet volume | PD is being correlated with a change in platelet volume. PD patients usually have an increased platelet volume compared to both AD patients as well as control individuals. | (Koçer et al., 2013; Ferrer-Raventós and Beyer, 2021) |
| | Mean platelet volume (MPV) | MPV demonstrates the size of platelets. MPV was significantly elevated in PD patients as compared with healthy individuals, and an absence of α-synuclein leads to lower MPV. | (Koçer et al., 2013; Geyik et al., 2016; Tashkandi et al., 2018) |
| Platelet functional impairment | Adhesion | Platelet adhesion is the first and pivotal platelet functional parameter. Defective platelet adhesion was observed at the site of blood-vessel injury leading to a defective thrombus formation in α-synuclein deficient mice. | (Reheman et al., 2010) |
| | Activation | Platelet hyperactivity, and increased spreading, are observed in PD patients in comparison to healthy controls. But at the same time, a reduced platelet activation has been described in PD. | (Adams et al., 2019) |
| | Aggregation | Platelet aggregation induced by ADP and epinephrine was observed to be significantly reduced in PD patients. Both MPP+, a PD inducing compound as well as α-synuclein also inhibit platelet aggregation. | (Sharma et al., 1991; Lim et al., 2009; Pontarollo et al., 2021) |
| | Degranulation | Exogenous α-synuclein inhibits thrombin- or ionomycin-induced alpha-granule release from platelets but remains ineffective for dense-granule and lysosomal-granule. A low amount of α-granules are reported to be released from platelets in PD patients. | (Park et al., 2002; Simonidze and Samushia, 2014) |
| Platelet organelle dysfunctions | Plasma membrane | Platelet membrane fluidity remains unchanged in PD patients, but the membrane and adjacent areas are essential for the localization of α-synuclein. | (Factor et al., 1994; Hashimoto et al., 1997; Park et al., 2002) |
| | Mitochondrial system | Platelet mitochondrial is the primary vulnerable organelle in PD patients, where reduced respiration rate, and ATP production occurs due to decreased activity of platelet mitochondrial complex-I, II, III as well as coenzyme Q10. | (Haas et al., 1995; Behari and Shrivastava, 2013; Ferrer-Raventós and Beyer, 2021; Clark et al., 2021) |
| | Open canalicular system (OCS) | OCS is the primary route of transport of cargo and vesicular trafficking in platelets, where platelet vacuoles, the part of OCS were observed to be larger in PD. | (Factor et al., 1994) |
| | Ubiquitin proteasomal system (UPS) | Alteration of platelet UPS is controversial, but parkin, an E3 ubiquitin ligase as well as a PD biomarker is highly expressed in blood platelets as compared to other tissues, which helps in mitophagy. | (Lee et al., 2020) |
| Platelet storage & secretory components | Alpha-synuclein (α-syn) | Platelets contain the highest amount of α-synuclein per milligram of cellular protein in the blood. It stores these proteins inside the cytoplasm and other compartments. It is also found to be secreted in platelet extracellular vesicles into circulation. | (Park et al., 2002; Tashkandi et al., 2018; Pei and Maitta, 2019; Stefaniuk et al., 2021) |
| | Dopamine (DA) & Dopamine transporter (DAT) | Platelets store about 99% of the circulatory dopamine in their dense-granules and secrete it upon activation. Platelets also express dopamine receptors as well as the transporter, DAT. A decreased dopamine uptake by platelet storage granules is observed in PD. | (Rabey et al., 1993; Osinga et al., 2016; Beura et al., 2022) |
| | Glutamate (Glu) & Glutamate receptor (GlutR) | Platelets express not only glutamate in their dense-granules but also the glutamate receptors on their surfaces. Although an increased level of glutamate is found in platelets of PD patients, the glutamate uptake remains reduced in PD platelets. | (Ferrarese et al., 1999; Gautam et al., 2019) |
| | γ-Aminobutyric acid (GABA) | PD is associated with a decreased expression of benzodiazepine receptors, a part of the GABA receptor complex present on the outer mitochondrial membrane of the platelet. | (Bonuccelli et al., 1991) |
| | Monoamine oxidase B (MAO-B) | MAO-B mediates the production of ROS as a side product *via* oxidative deamination of several cellular molecules. The platelet MAO-B activity is reported to be significantly higher in patients with PD. | (Bonuccelli et al., 1990; Danielczyk et al., 1988; Zhou et al., 2001) |
| | Tyrosine hydroxylase (TH) | TH is the crucial rate-limiting enzyme that catalyzes the conversion of tyrosine into L-DOPA in the dopamine biosynthesis pathway. Platelets are also reported to be equipped with TH. | (Drozdov and Anokhina, 1990) |
| | Parkin (PRKN) | Parkin is an essential E3 ubiquitin ligase that helps in mitophagy and inhibits apoptosis. Parkin is highly expressed in blood platelets. | (Lee et al., 2020) |
| | Vesicular monoamine transporter 2 (VMAT2) | VMAT2 helps in the cytoplasmic uptake of monoamine neurotransmitters like dopamine and serotonin into dense-granules of the platelet. VMAT2 mRNA levels are reported to be reduced in platelets in PD patients. | (Sala et al., 2010) |
| | Vacuolar protein sorting-associated protein 35 (VPS35) | VPS35 helps in cargo transport from Golgi and helps in autophagy. Platelet lysosomal-granules are reported to carry VPS35. | (Rijkers et al., 2017) |
| | PTEN-induced kinase 1 (PINK1) | PINK1 is a mitochondrial kinase that helps in protecting mitochondrial stress-induced apoptosis. It along with parkin helps in mitophagy. PINK1 deficiency causes increased platelet activation, suggesting the theory of PINK1-driven mitophagy in alleviating platelet stress. | (Walsh et al., 2018) |

(*continued on next page*)

**Table 1** (*continued*)

| Category | Parameters/Factors | Contributions in PD | Refs. |
|---|---|---|---|
| | Lysosomal associated membrane protein-2 (LAMP-2) | LAMP2 is a lysosomal membrane glycoprotein, that is confirmed to be present both on lysosomal-granules and dense-granules of the platelet. It is an activation-dependent platelet surface glycoprotein. | (Silverstein and Febbraio, 1992; Israels et al., 1996) |
| | Deglycase 1 (DJ1) | DJ1 is an oxidative stress sensor that helps in the proper folding of α-synuclein and further inhibits its aggregation. Platelet is reported to store and express DJ1. | (Shi et al., 2010; Goelz et al., 2021) |
| | Leucine-rich repeat kinase 2 (LRRK2) | LRRK2 is an essential cellular protein that helps in mitochondrial functioning, vesicle trafficking, endocytosis, and autophagy. The proteomics experiments did not detect any LRRK2 in platelets. | (Babur et al., 2020) |
| | Antioxidants (SOD & GPx) | Antioxidants help in scavenging the excess ROS produced in the cell during PD. Reduced activity of SOD and GPx is observed in PD platelets explaining a possibility of increased ROS in platelets like neurons in the brain. | (Behari and Shrivastava, 2013; De La Fourniere et al., 2000) |

**Table 2**
**Diagnostic & therapeutic applications of platelet in Parkinson's disease:** A detailed description of the potential PD-specific biomarkers for early diagnosis, along with the therapeutic effects of different antiplatelet drugs, and platelet-secreted growth factors ameliorating PD.

| Category | Subject | Findings | | Ref. |
|---|---|---|---|---|
| **Diagnosis** | Platelet proteins & RNAs for PD diagnosis | Platelet extracellular vesicles (EVs) contribute about 70–90 % of all circulating EVs. As α-synuclein is found in platelet EVs, the concentration of α-synuclein in platelet EVs may function as a potential diagnostic biomarker for PD. | | (Teixeira et al., 2016; Stuendl et al., 2021) |
| | | The peripheral blood platelets are rich in *WASH5P, MALAT1, EEF1A1* , and *CTSS*, which may contribute to being promising biomarkers for an effective and accurate diagnosis for PD in the future. | | (Zhang et al., 2021) |
| | | Elevated blood expression of the platelet-related miRNAs miR-22-3p, miR-146a and miR-155 are found to be plausible biomarkers for an early diagnosis of PD. | | (Espinosa-Parrilla et al., 2019) |
| | | Attenuated blood expression of the platelet-related miRNAs like hsa-miR-128-3p and hsa-miR-139-5p are potential biomarkers for the early detection of PD. | | (Gámez-Valero et al., 2021) |
| **Therapeutics** | Antiplatelet drugs for PD treatment | Aspirin | Aspirin inhibits platelet COX enzymes, thus inhibiting the $TXA_2$-mediated platelet aggregation as well as platelet-regulated secretion of ROS. It is being reported to inhibit mitochondrial dysfunction, scavenge ROS besides improving the movement impairment in PD conditions. | (Majithia and Bhatt, 2019; Thrash-Williams et al., 2016; Fyfe, 2020) |
| | | Ticagrelor | Ticagrelor inhibits platelet *via* non-competitive as well as reversible inhibition of platelet P2Y12 receptor. It inhibits neuroinflammation and improves cerebral blood flow, and exhibits neuroprotection. | (Michelson, 2010; Yamauchi et al., 2017; Li et al., 2022) |
| | | Cilostazol | Cilostazol inhibits platelet aggregation *via* the cAMP-PKA mediated signaling pathway. It upregulates the Nurr1 expression, as well as SIRT-1/autophagy, and GSK-3β/apoptosis cross-regulation in PD rats. | (Sun et al., 2002; Hedya et al., 2018) |
| | | Celastrol | Celastrol potentially inhibits ADP and thrombin-induced adhesion, activation, and aggregation of the platelet. It inhibits mitochondrial damage, as well as promotes the reduction of neuroinflammation besides suppressing motor impairment in PD. | (Hu et al., 2009; Lin et al., 2019) |
| | | Prasugrel | Prasugrel inhibits platelet *via* irreversible and competitive inhibition of platelet P2Y12 receptor. It potentially reduces oxidative stress, abates neuroinflammation, activates neurogenesis, and improves memory functioning. | (Majithia and Bhatt, 2019; Ohno et al., 2016; Gomaa et al., 2021) |
| | | Nobiletin | Nobiletin inhibits platelet aggregation *via* inhibition of PLCγ2-PKC and Akt/MAPK pathways. It also rescues motor and cognitive dysfunctions in PD mice by inhibiting glial activation and subsequent neuroinflammation and further enhances the secretion of dopamine in the striatum. | (Jayakumar et al., 2017; Yabuki et al., 2014; Jeong et al., 2015) |
| | | Quercetin | Quercetin inhibits platelet *via* inhibiting COX-1 signaling and the cAMP/PLA2 pathway. It decreases α-synuclein aggregation, increases antioxidant status, and reduces the expression of neuroinflammatory molecules in PD. | (Beura et al., 2021; Tamtaji et al., 2020; Madiha et al., 2021) |
| | | Dipyridamole | Dipyridamole is a PDE inhibitor that inhibits the adhesion and aggregation of platelet in a NO-mediated pathway. It can attenuate oxidative stress and promotes the expression of TH in PD. | (Kim and Liao, 2008; Rodríguez-Pallares et al., 2002; Höllerhage et al., 2017) |
| | | 3-*N*-butylphthalide | 3-*n*-butylphthalide reduces platelet activation *via* inhibition of cPLA2-mediated TXA2 synthesis. It suppresses oxidative stress, mitochondrial damage, α-synuclein aggregation besides augmenting neuronal antioxidant enzymes in PD. | (Ye et al., 2015; Huang et al., 2010; Chen et al., 2018) |
| | | Mitoquionone | Mitoquinone inhibits adhesion, activation, and aggregation of platelets by decreasing the secretion of ATP induced by TRAP-6. It can reduce the generation of ROS, inhibits mitochondrial damage, and apoptosis, thus protecting DA neurons in PD. | (Méndez et al., 2020; Méndez; Xi et al., 2018; Kim et al., 2020) |
| | Platelet-secreted growth factors for PD treatment | Endoret containing platelet-secreted growth factors resulted in a decreased neuroinflammation, and improved motor performance by the significant reduction in NF-κB activation, NO, COX-2, and TNF-α expression in the substantia nigra of PD mice. | | (Anitua et al., 2015) |
| | | Heat-treated platelet pellet lysates prevented MPP + induced LUHMES neuronal death by downregulating neuroinflammation in PD mice via inhibiting COX-2. | | (Chou et al., 2017) |
| | | Heat-treated human platelet lysates enhanced the expression of TH and neuron-specific enolase (NSE) in LUHMES cells and rescued them from neuronal death. | | (Nebie et al., 2019) |
| | | The human platelet lysate prevents MPP + induced LUHMES neuronal death by activating specific signaling cascade *i.e.* the Akt pathway and, to a smaller extent, the MEK pathway. | | (Gouel et al., 2017) |

*5.2.1.1. Aspirin.* Aspirin (acetylsalicylic acid) is an excellent option targeting both platelet and neuron as it irreversibly inhibits the platelet COX enzyme, thus inhibiting the synthesis of $TXA_2$, which causes platelet aggregation and vasoconstriction. Moreover, small doses of aspirin are required for COX-1 blockade, and high doses can inhibit COX-2 (Majithia and Bhatt, 2019). A research finding by Thrash et al. demonstrated that salicylate significantly reduced methamphetamine-induced DA neuronal death by inhibiting mitochondrial dysfunction and scavenged ROS, besides improving the movement impairment in PD conditions (Thrash-Williams et al., 2016). In recent research, Fyfe et al. also suggested the potential role of aspirin as a therapeutic option for reducing the risk of LRRK2-manifested PD (Fyfe, 2020).

*5.2.1.2. Ticagrelor.* Ticagrelor is a potent antiplatelet drug that functions as a non-competitive, reversible cyclopentyl triazolopyrimidine P2Y12 receptor antagonist (Michelson, 2010). In a recent study by Yamauchi et al., it is established that ticagrelor inhibits neuroinflammation by attenuating microglial activation, apart from improving cerebral blood flow and neuroprotection *via* phosphorylation of eNOS and ERK1/2 (Yamauchi et al., 2017). As stroke is considered a prominent vascular risk factor for the development of PD, ticagrelor is proposed to reduce the risk of vascular complications, including stroke as well as dementia in PD patients (Li et al., 2022). Hence, ticagrelor could be a novel therapeutic candidate for the treatment of PD.

*5.2.1.3. Cilostazol.* Cilostazol is a potent antiplatelet drug that selectively inhibits phosphodiesterase 3 (PDE3), thus inducing intracellular concentration of cAMP, and further regulates cAMP/PKA-mediated inhibition of platelet aggregation as well as activation (Sun et al., 2002). In a study, Hedya et al. confirmed that cilostazol could attenuate neuroinflammation *via* inhibiting GSK-3β-mediated activation of proinflammatory molecules like Nf-κβ, TNF-α, and IL-1β. They also proved that cilostazol could potentially upregulate Nurr1 expression, resulting in the preservation of DA neurons, as well as improving the motor activity along with an increase in striatal TH content in PD rats (Hedya et al., 2018). This study suggests that cilostazol could be a potential drug solution for the therapeutic treatment of PD.

*5.2.1.4. Celastrol.* Celastrol is an antiplatelet drug that inhibits ADP-induced expression of P-selectin (CD62p), a platelet activation marker, and integrin αIIbβ3 activation, thus affecting platelet adhesion as well as platelet aggregation (Hu et al., 2009). In a recent study, Lin et al. confirmed that celastrol ameliorates autophagy impairment by upregulating autophagosome proteins like Atg7, Atg12, LC-II, Beclin1, and VPS34, in addition to promoting mitophagy by upregulating the expression of PINK1 and DJ-1. The same group also proved that celastrol triggers the rescue of $MPP^+$-induced death of DA neurons by inhibiting mitochondrial damage, as well as a reduction in neuroinflammation, besides suppressing motor impairment in the PD model (Lin et al., 2019). Therefore, celastrol could be a potential medication for the therapeutic treatment of PD.

*5.2.1.5. Prasugrel.* Prasugrel is another antiplatelet drug that functions as an irreversible, competitive, thienopyridine P2Y12 receptor antagonist (Majithia and Bhatt, 2019). In a recent study by Gomaa et al., it is reported that prasugrel can potentially reduce oxidative stress, abate neuroinflammation, activate neurogenesis, and improve memory functioning by activating the cAMP/PKA/CREB/BDNF pathway in mouse models (Gomaa et al., 2021). Gait impairment is considered a prominent motor symptom of PD, and Ohno et al., reported that prasugrel improved abnormal gait by improving cerebral blood flow, suggesting it to be used as a potential therapeutic candidate for PD (Ohno et al., 2016).

*5.2.1.6. Nobiletin.* Nobiletin is an antiplatelet flavone that can efficiently reduce platelet aggregation *via* inhibition of PLCγ2-PKC and Akt/MAPK pathways (Jayakumar et al., 2017). In a study, Jeong et al. confirmed that nobiletin administration into $MPP^+$-induced PD rats rescued not only DA neurons of substantia nigra, but also inhibited microglial activation and subsequently preserved the expression of glia-cell derived neurotrophic factor (Jeong et al., 2015). Another study by Yabuki et al. also observed a similar effect of nobiletin, where it restored the CaMKII-cAMP-PKA-dependent TH level and maintained dopamine concentration as well as calcium mobilization in the striatal area and hippocampal CA1 region (Yabuki et al., 2014). Therefore, nobiletin could be a suitable candidate for the therapeutic treatment of PD.

*5.2.1.7. Quercetin.* Quercetin is a natural flavonoid and a potent platelet antagonist that inhibits platelet *via* regulation of the cAMP/PLA2 pathway and downregulation of intracellular $Ca^{2+}$ mobilization, IL-6, IL-1β, in addition to an inhibition of COX-1 (Beura et al., 2021). Tamtaji et al. discussed the neuroprotective effects of quercetin, which include the decrease in α-synuclein aggregation, amelioration of impaired autophagy and antioxidant status, and a reduced expression of neuroinflammatory molecules like TNF-α, IL-6, and IL-1β in PD models (Tamtaji et al., 2020). In a recent study, Madiha et al. reported that quercetin could attenuate oxidative stress and significantly improves motor and cognitive functions by restoring the dopamine level in rotenone-induced PD rats, thereby proposing it to be used as a potential therapeutic option for the treatment of PD (Madiha et al., 2021).

*5.2.1.8. Dipyridamole.* Dipyridamole is an effective inhibitor of cGMP-dependent PDE and an adenosine transporter. In addition to this, the same compound can also reduce platelet adhesion as well as platelet aggregation in a NO-mediated pathway (Kim and Liao, 2008). In a study, Rodríguez-Pallares et al., confirmed that dipyridamole could attenuate oxidative stress and promoted the expression of TH, an essential enzyme for dopamine biosynthesis. It also triggered the neurogenesis of DA neurons from EGF-responsive mesencephalic precursors (Rodríguez-Pallares et al., 2002). In a recent report, Höllerhage et al. reported that dipyridamole could rescue from neuronal death by reducing the release of lactate dehydrogenase (LDH) from α-synuclein overexpressing LUHMES neurons, suggesting the potential effect of this drug as a therapeutic candidate for PD treatment (Höllerhage et al., 2017).

*5.2.1.9. 3-N-butylphthalide.* 3-*N*-butylphthalide (NBP) is a potent antiplatelet drug that inhibits platelet activation *via* inhibition of cPLA2-mediated production of $TXA_2$ as well as PDE (Ye et al., 2015). In a study by Huang et al., NBP was reported to suppress oxidative stress, mitochondrial damage, and α-synuclein accumulation, besides augmenting neuronal antioxidant GSH level in an $MPP^+$-induced cellular model of PD (Huang et al., 2010). In a later study, Chen et al. demonstrated that NBP could reduce the neurotoxicity in an LPS-induced PD mouse by promoting the dopaminergic expression of TH, thus decreasing the nuclear deposition of α-synuclein, mitigating microglia activation, in addition to the improvement in PD symptoms, like dyskinesia (Chen et al., 2018). These findings propose NBP to be used as a potential drug candidate for the management of PD.

*5.2.1.10. Mitoquinone.* Mitoquinone (MitoQ) is an antiplatelet drug that inhibits the adhesion, activation, and aggregation of human platelets by decreasing the secretion of ATP induced by thrombin receptor activator peptide-6 (TRAP-6) (Méndez et al., 2020). In a finding, Xi et al. reported that MitoQ could reduce the generation of ROS and could potentially inhibit mitochondrial damage and neuronal apoptosis. It protects DA neurons by activating PGC-1α to promote MFN2-dependent mitochondrial fusion in the 6-OHDA-induced PD model (Xi et al., 2018).

In a later report, Kim et al. confirmed that MitoQ could improve motor function by ameliorating mitochondrial functions, thus reducing the death of DA neurons in the $MPP^{+}$-induced mice model for PD (Kim et al., 2020). This suggests the potential efficacy of MitoQ to be used as a possible therapeutic drug candidate for PD.

#### 5.2.2. Platelet growth factors as therapeutic candidates for Parkinson's disease

Besides the diagnosis aspect, blood platelets have also been used for a long time in wound healing and tissue engineering owing to their unique collection of growth factors such as brain-derived neurotrophic factors (BDNF), epidermal growth factor (EGF), platelet-derived growth factor (PDGF), vascular endothelial growth factor (VEGF), basic fibroblast growth factor (bFGF), and insulin-like growth factor (IGF) as well as regulators of synaptogenesis like reelin and ephrins (Fig. 3) (Beura et al., 2022). In one study, it is found that platelet-rich plasma (PRP) promotes axon growth by potentiating the improvement of oligodendrogenesis, remyelination, and functional abilities in the multiple sclerosis (MS) mouse model. The PRP treatment subsequently attenuated the number of astrocytes, microglia, and infiltrating inflammatory cells, as well as the expression of proinflammatory cytokines (Borhani-Haghighi and Mohamadi, 2019). Several studies have also reported that Endoret-containing platelet-secreted growth factors can enhance the activation of neuronal progenitor cells, enhancing the hippocampal neurogenesis process, hence reducing the Aβ-induced neurodegeneration in an AD mouse model (Anitua et al., 2013). These platelet-derived growth factors can modulate astrocyte activation, reducing inflammatory responses, and promoting Aβ degradation, in addition to the symptomatic improvements in anxiety, learning, and memory in AD mice (Anitua et al., 2014). Similar to the above findings in AD, Endoret-containing platelet-secreted growth factors also resulted in mitigating inflammatory responses and improved motor performance by the robust reduction in neuronal NF-κB, NO, COX-2, and TNF-α expression in the substantia nigra of PD mice (Anitua et al., 2015). Furthermore, Gouel et al. also demonstrated that human platelet lysates are a rich source of various growth factors, which can potentially promote neuronal growth and proliferation. In the study, human platelet lysate prevented $MPP^{+}$-induced LUHMES (an *in vitro* PD model) neuronal death by activating specific pathways, *i.e.,* the Akt pathway and, to a lesser extent, the MEK pathway (Gouel et al., 2017). Similarly, Chou et al. have also reported that human platelet lysate is neuroprotective in nature, where these lysates containing platelet-rich growth factors can rescue neuroinflammation in PD mice by inhibiting COX-2 in $MPP^{+}$-induced LUHMES cell lines (Chou et al., 2017). Another study by Nebie et al. reported that heat-treated human platelet lysates enhanced the expression of neuronal TH, which is typically downregulated in PD patients, and neuron-specific enolase (NSE) in $MPP^{+}$-induced LUHMES cells, and further rescued them from neuronal death (Table 2) (Nebie et al., 2019). These findings suggest that platelet and its secretory growth factors can modulate neurogenesis and can potentially help in restoring DA neurons of SNpc regions in PD.

## 6. Challenges and future perspectives

Although there has been a substantial advancement in deducing the structural, functional as well as secretory alterations of platelet in PD, there are several vital subcellular mechanisms remain to be discovered in the near future. Since there are limited studies available regarding the discovery of crucial PD-specific biomarkers, still these findings are consolatory for future research in this area. The established preventive role of several antiplatelet drugs is admirable, but the lack of clinical breakthroughs in the treatment procedure for PD remains to be disputable, as a majority of the data regarding the efficacy of antiplatelet drugs in PD come from laboratory studies. The antiplatelet drugs could potentially be the add-on medications or co-treatments that can be used as adjuvants with established classical drugs to increase the efficacy of already existing drugs for better PD management. The most obvious challenge in the application of single or dual antiplatelet regimens is the magnitude of inhibition of platelet, as platelet is indispensable for the growth and proliferation of damaged tissues. Platelet is a repository of various growth factors, which can enable faster neuroregeneration; hence complete platelet inhibition should not be recommended. The most critical aspect will be the signaling *via* which platelet releases its granular growth factors and the signaling through which antiplatelet drugs elicit their mode of action (Fig. 4.). Moreover, the prolonged use of these antiplatelet drugs could potentially cause the intracerebral bleeding issue, as seen in several diseases, which needs to be optimized in a controlled manner. The platelets are also a key source of several immunomodulatory molecules that could potentially trigger inflammation. Therefore, the formulated platelet lysate could induce a severe immune response in the recipient patient if not radically controlled. Hence, the platelet factors should be contamination-free along with having a low immunogenicity property to be considered the successful candidates for PD treatment. The local controlled release of antiplatelet drugs and platelet-rich growth factors needs special attention from nanotechnology, where fabricated nanomaterials are better options in the future, which could be proven as a requisite for specific drug delivery vehicles as well as formulated bio-scaffolds for platelet-rich growth factors for the treatment of PD.

## 7. Conclusion

Due to an increased number of aging-related diseases and concomitantly elevated prevalence of PD in the world population, the need for its efficient management is requisite. Nevertheless, there is no substitute for the precise diagnosis of definite PD. However, recent findings suggest that circulatory blood platelets, due to their significant similarities with DA neurons, could potentially provide a valuable source of efficient biomarkers to assist in early PD detection. Furthermore, these tiny platelets could be promising peripheral surrogates to detect several fundamental molecular changes related to PD, which could provide crucial information for the development of theranostic tools. As thrombosis is an associated major risk factor for PD, we have proposed several essential antiplatelet drugs with established findings to be used as the adjuvants along with the frontline drugs for the better management of PD. Moreover, the platelet-rich growth factors could be the pragmatic biotherapeutic approach to improve the PD severity by not only promoting neurogenesis but also attenuating neurotoxicity (Table 2). Further research will elucidate the role of platelet as a peripheral model for PD and unveil more about the diagnostic as well as the therapeutic effects of platelet and their growth factors in PD.

## CRediT authorship contribution statement

S.K.B. and S.K.S designed the manuscript. S.K.B. wrote the manuscript. A.R.P. and P.Y. prepared the illustrated figures and descriptive tables. All authors read and approved the final manuscript.

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## Competing interests

The authors declare no potential competing interests.

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